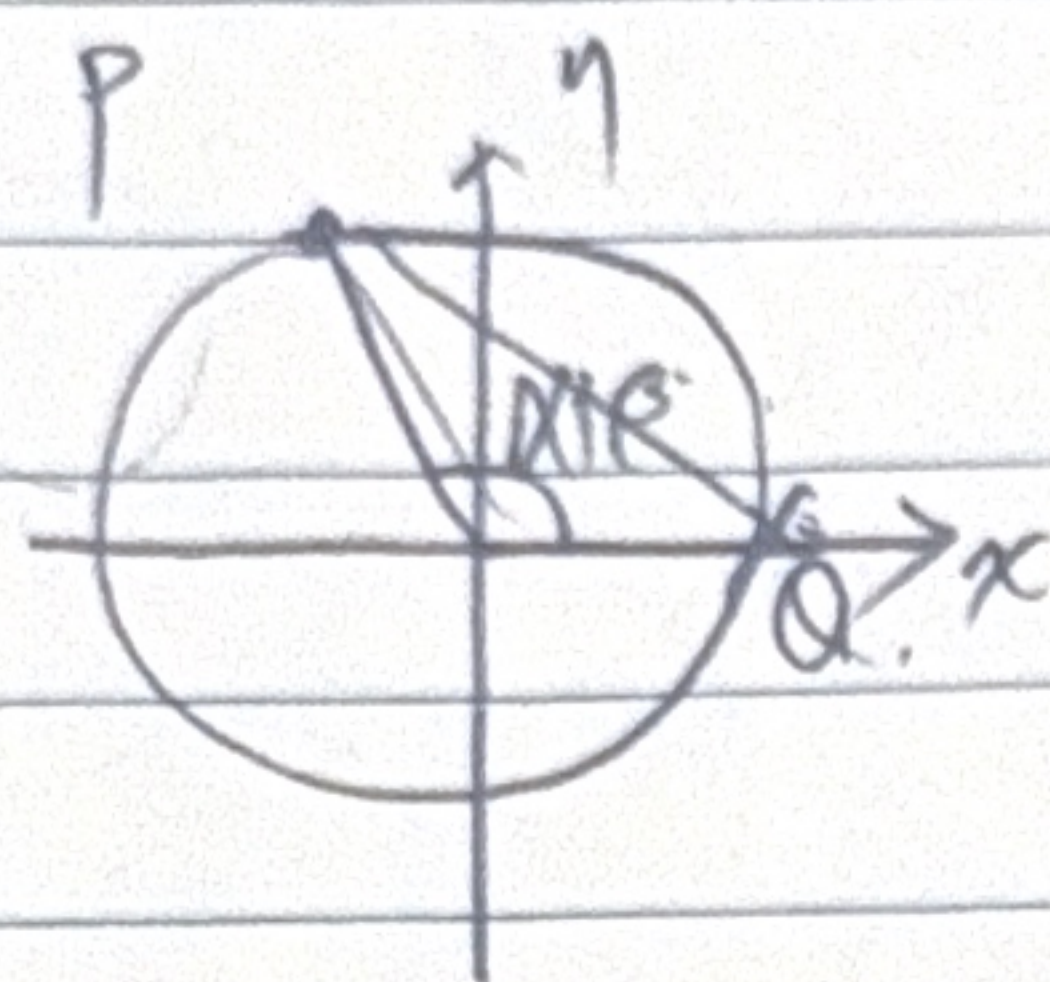




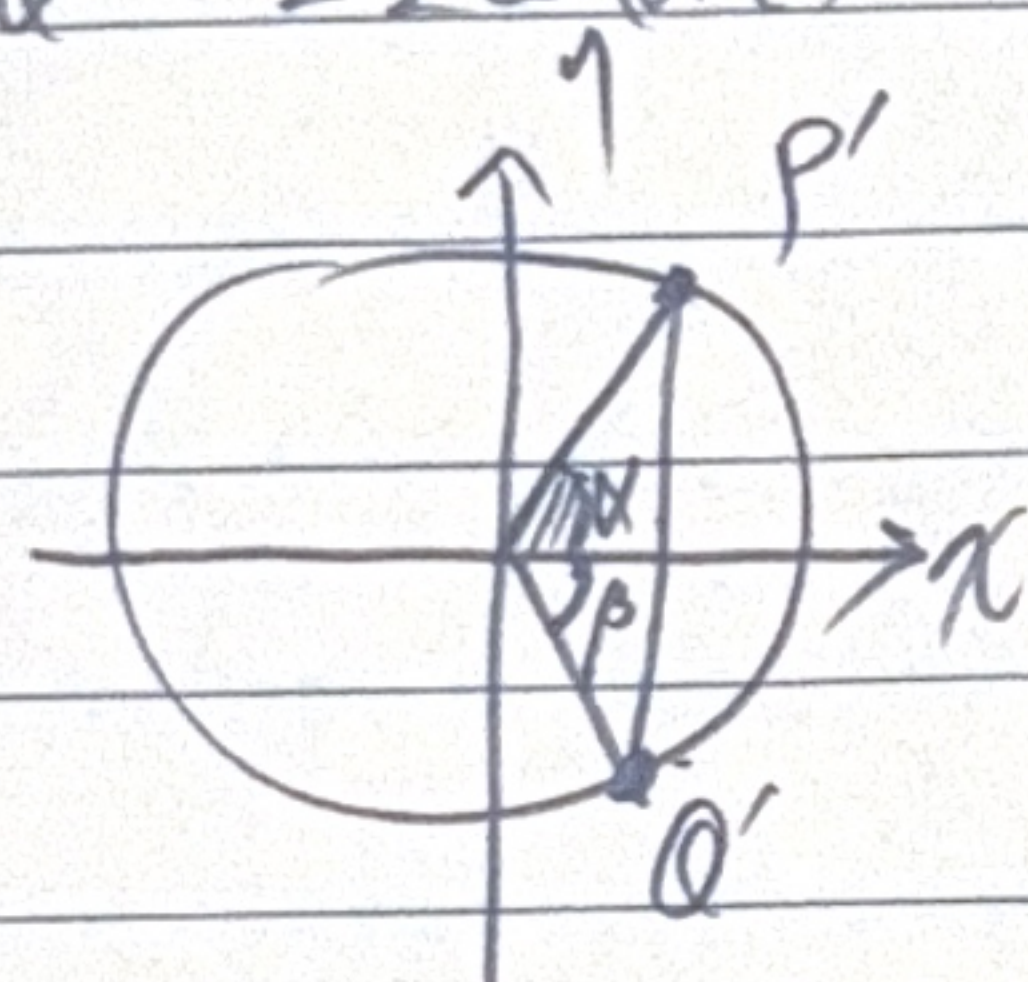
$$P(\cos(\alpha), \sin(\alpha))$$

$$Q(1, 0)$$

$$\overline{PQ}^2 = \overset{\Delta x}{|1 - \cos(\alpha)|^2} + \overset{\Delta y}{|0 - \sin(\alpha)|^2}$$

$$= 2 - 2\cos(\alpha)$$

$$\therefore \overline{PQ} = 2 - 2\cos(\alpha)$$



$$P'(\cos \alpha, \sin \alpha)$$

$$Q'(\cos(-\beta), \sin(-\beta))$$

$$\overline{P'Q'}^2 = \overset{\Delta x}{|\cos(-\beta) - \cos \alpha|^2} + \overset{\Delta y}{|\sin(-\beta) + \sin \alpha|^2}$$

$$= 2 - 2\cos \alpha \cos(-\beta) - 2\sin \alpha \sin(-\beta)$$

$$= 2 - 2\cos \alpha \cos \beta + 2\sin \alpha \sin \beta$$

$$\therefore \overline{P'Q'}^2 = 2 - 2\cos \alpha \cos \beta + 2\sin \alpha \sin \beta$$

$$\text{let } \beta \text{ be } -\beta \begin{cases} \sin(-\beta) = -\sin \beta \\ \cos(-\beta) = \cos \beta \end{cases}$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta - \sin \alpha \sin(-\beta)$$

$$= \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\checkmark \overline{PQ}^2 = \overline{P'Q'}^2$$

$$\Rightarrow 2 - 2\cos(\alpha) = 2 - 2\cos \alpha \cos \beta + 2\sin \alpha \sin \beta$$

$$= -2\cos(\alpha) = -2\cos \alpha \cos \beta + 2\sin \alpha \sin \beta$$

$$= \cos(\alpha) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\therefore \cos(\alpha - \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\boxed{\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta}$$